

9.1.22.

- a. How many positive three-digit integers are multiples of 6?
- b. What is the probability that a randomly chosen positive three-digit integer is a multiple of 6?
- c. What is the probability that a randomly chosen positive three-digit integer is a multiple of 7?

- a. How many positive three-digit integers are multiples of 6?

Let $6k$ be the multiples of 6, for some $k \in \mathbb{Z}$, then there're as many multiples of 6 as there are integers from 17 to 166 inclusive, since:

$$6(16) = 96$$

$$6(17) = 102$$

$$6(166) = 996$$

$$6(167) = 1002$$

By theorem of the number of elements in a list, there are $166 - 17 + 1$, or 150, such integers. Hence, there are 150 three-digit integers that are divisible by 6.

Solution: a. There are 150 three-digit integers that are multiples of 6.

- b. What is the probability that a randomly chosen positive three-digit integer is a multiple of 6?

Let S be the set of the total number of integers from 100 through 999, which's $999 - 100 + 1 = 900$, by theorem of the number of elements in a list, and E_0 be the set of the total number of three-digit integers which're multiples of 6, and $E_0 \subseteq S$, and $N(S) = 900$, then $P(E_0)$ is the probability of a randomly chosen integer being a multiple of 6. Hence:

$$N(E_0) = 150$$

$$N(S) = 900$$

$$P(E_0) = \frac{N(E_0)}{N(S)} = \frac{150}{900} = \frac{1}{6} = 0.1666666667 \approx 0.17$$

Therefore, the probability of a randomly chosen integer being a multiple of 6 is 17%.

Solution: b. The probability of a randomly chosen integer being a multiple of 6 is 17%.

c. What is the probability that a randomly chosen positive three-digit integer is a multiple of 7?

Let $7k$ be the multiples of 7, for some $k \in \mathbb{Z}$, then there're as many multiples of 7 as there are integers from 15 to 142 inclusive, since:

$$7(14) = 98$$

$$7(15) = 105$$

$$7(142) = 994$$

$$7(143) = 1001$$

By theorem of the number of elements in a list, there are $142 - 15 + 1$, or 128, such integers. Hence, there are 128 three-digit integers that are divisible by 7. Let E_1 be the set of three-digit integers which're multiples of 7, and $E_1 \subseteq S$, and $N(S) = 900$, then $P(E_1)$ is the probability of a randomly chosen integer being a multiple of 7. Hence:

$$N(E_1) = 128$$

$$N(S) = 900$$

$$P(E_1) = \frac{N(E_1)}{N(S)} = \frac{128}{900} = \frac{32}{225} = 0.14\overline{22} \approx 0.14$$

Therefore, the probability of a randomly chosen integer being a multiple of 7 is 14%.

Solution: c. The probability of a randomly chosen integer being a multiple of 7 is 14%.